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Refractive index-structure correlations in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses

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Abstract

The current structural models have been used to analyse the refractive index data of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses ($\text{Al}_2\text{O}_3/\text{Na}_2\text{O} \leq 1$). The SiO_2 content is the sole factor that controls the refractive index. Values could be obtained for the factors with which each structural unit contributes in the refractive index. The content of Al_2O_3 or Na_2O has no effect on the refractive index. The factors (differential refraction) are constant and do not change with composition. They have the same values for $\text{Na}_2\text{O}-\text{SiO}_2$ glasses. The differential refraction of a structural unit increases linearly with increasing the number of non-bridging oxygen ions. The difference of the contribution to the refractive index from a silicate unit to the next equals to a half of that for AlO_4 tetrahedron. The effect could be attributed to the change in both the concentration and differential refraction of structural units. The obtained factors for the structural units are useful in calculating the refractive index with a high degree of accuracy. © 2002 Elsevier Science B.V. All rights reserved.

1. Introduction

Aluminum oxide plays a distinguished role in the structure of glass. It belongs to the group of intermediate oxides. Therefore it is necessary to melt this oxide with a network forming oxide to form a glass. In $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{B}_2\text{O}_3$ glasses both Al_2O_3 and B_2O_3 associate, at the same time, with alkali oxide regardless the glass composition [1]. Some of Al_2O_3 would be present in the form of AlO_4 units and the rest exists either as AlO_6 units or in the form of triclusters. The distribution of

Na_2O between Al_2O_3 and B_2O_3 depends at the first place on the glass composition.

The situation differs to a great extent in the case of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses. In glasses having $\text{Al}_2\text{O}_3 > \text{Na}_2\text{O}$ triclusters would form [2]. For $\text{Al}_2\text{O}_3/\text{Na}_2\text{O} \leq 1$ aluminum oxide associates with an equivalent quantity of Na_2O to form AlO_4 units at a rate of two units per Al_2O_3 molecule [3,4]. The rest of Na_2O incorporates into the silicate matrix and form SiO_4 tetrahedra with non-bridging oxygen ions (NBOs). The number of NBOs per SiO_4 tetrahedron depends on the glass composition [5]. For $(\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 0.5$ Q_4 units convert into Q_3 ones. A Q_4 unit is a SiO_4 tetrahedron without NBOs and a Q_3 unit is a SiO_4 tetrahedron having one NBO. For $0.5 < (\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 1$ Q_3 units would convert into Q_2 units. The

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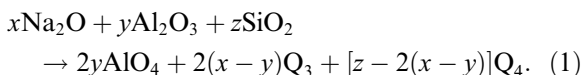
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latter are SiO_4 tetrahedron having two NBOs per unit.

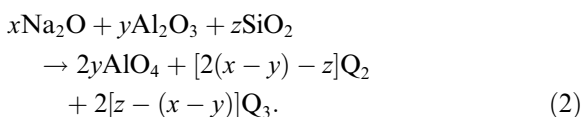
The above concepts were used to analyze the density data of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses having $\text{Al}_2\text{O}_3/\text{Na}_2\text{O} \leq 1$ [5]. It was indicated that the volume of each structural unit is constant and does not depend on the glass composition. In the present work an analyses will be carried out for the refractive index data of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses having $\text{Al}_2\text{O}_3/\text{Na}_2\text{O} \leq 1$. The aim is to estimate the contribution of each structural unit in the refractive index and to correlate the property with the glass structure.

2. Procedure

In $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses Al_2O_3 has priority to associate itself with Na_2O to form AlO_4 units. Therefore the formation of structural units in a $x\text{Na}_2\text{O} \cdot y\text{Al}_2\text{O}_3 \cdot z\text{SiO}_2$ glass, for $(\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 0.5$, would be [5]



In glasses with $0.5 < (\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 1$, Q_4 units do not more exist and Q_2 units are formed from Q_3 units. Formation of the structural units in this case goes on as



In previous works the refractive index (μ) was correlated with structure of alkali [6] and alkaline earth [7] silicate glasses. The refractive index could be represented by the additive relation

$$\mu = \sum n_u f_u. \quad (3)$$

In this relation n_u represents the number of the structural unit per mole of glass and f_u is a dimensionless factor called the 'differential refraction' of the structural unit.

On the basis of Eqs. (1) and (3) the refractive index of a $x\text{Na}_2\text{O} \cdot y\text{Al}_2\text{O}_3 \cdot z\text{SiO}_2$ glass can be given, for $(\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 0.5$, by

$$\mu = \{2yf_{\text{Al}} + 2(x-y)f_3 + [z-2(x-y)]f_4\}N_A. \quad (4a)$$

In this relation f_{Al} , f_4 and f_3 represent respectively the differential refraction for the AlO_4 , Q_4 and Q_3 units and N_A is the Avogadro's number. The quantity $2yN_A$ represents the number (n_{Al}) of AlO_4 units per mole of glass, $2(x-y)N_A = n_3$ the number of Q_3 units and $[z-2(x-y)]N_A = n_4$ the number of Q_4 units.

In the region $0.5 < (\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 1$ the refractive index is expressed as

$$\mu = \{2yf_{\text{Al}} + [2(x-y) - z]f_2 + 2[z - (x-y)]f_3\}N_A, \quad (4b)$$

where f_2 is differential refraction for the Q_2 unit and $2[z - (x-y)]N_A = n_2$ the number of Q_2 units per mole of glass.

Eqs. (4a) and (4b) can be treated to calculate f_{Al} , f_4 , f_3 and f_2 . This will be shown in Section 4.

3. Results

Fig. 1 represents the change in the determined refractive index (μ_D) with the SiO_2 content for $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses. The data (Table 1)

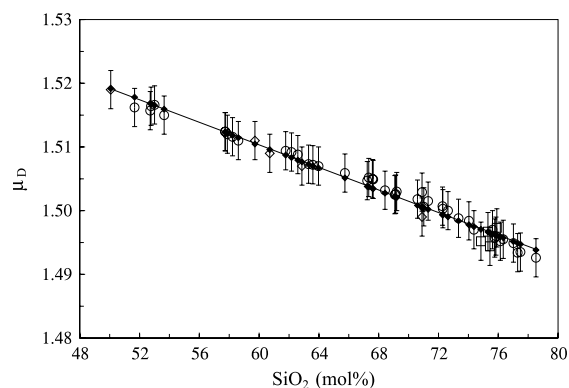


Fig. 1. Refractive index (μ_D) of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses as a function of the SiO_2 content. The data for determined refractive index (μ_D) are taken from Refs. [8,9] ($\diamond$), [10] ($\circ$) and [11] ($\square$). The solid line is the fitting plot of all the data. The ($\blacklozenge$) symbol represents calculated refractive index (μ_C). The μ_C values were calculated by using Eqs. (4a) and (4b) and the f_u values given in Table 2. The error limit in μ_D is estimated to be ± 0.003 .

Table 1

Determined and calculated refractive index for Na₂O–Al₂O₃–SiO₂ glasses: (panel A) μ_D values are taken from [9,10]; (panel B) μ_D values are taken from [8,9,11]

SiO ₂ (mol%)	Al ₂ O ₃ (mol%)	Na ₂ O (mol%)	μ_D	μ_C
<i>Panel A</i>				
51.664	1.731	46.605	1.5162	1.5178
52.769	4.305	42.926	1.5164	1.5168
53.006	4.121	42.873	1.5166	1.5165
53.645	5.945	40.410	1.5150	1.5159
52.716	2.976	44.308	1.5157	1.5168
58.220	6.105	35.675	1.5116	1.5118
57.871	4.260	37.869	1.5120	1.5121
58.609	6.007	35.384	1.5110	1.5114
57.716	2.944	39.340	1.5124	1.5123
57.742	2.942	39.316	1.5124	1.5123
62.586	2.954	34.459	1.5088	1.5079
62.170	1.701	36.129	1.5092	1.5083
61.774	0.653	37.573	1.5094	1.5087
65.741	0.610	33.649	1.5059	1.5051
63.596	6.101	30.303	1.5072	1.5070
63.975	5.939	30.086	1.5070	1.5067
63.304	4.020	32.676	1.5073	1.5073
68.418	6.169	25.412	1.5032	1.5028
67.628	2.882	29.490	1.5049	1.5035
69.075	5.827	25.098	1.5025	1.5022
67.335	1.728	30.937	1.5052	1.5037
67.234	1.449	31.317	1.5047	1.5038
67.574	1.218	31.207	1.5050	1.5035
69.135	2.689	28.175	1.5026	1.5021
69.206	1.776	29.018	1.5030	1.5021
70.598	4.124	25.278	1.5018	1.5008
71.013	3.511	25.476	1.5006	1.5005
70.900	0.579	28.521	1.5029	1.5006
71.310	1.217	27.473	1.5015	1.5002
72.290	2.945	24.765	1.5003	1.4993
72.639	2.982	24.379	1.5000	1.4990
72.265	1.713	26.021	1.5007	1.4994
73.340	3.958	22.702	1.4988	1.4984
74.371	5.747	19.882	1.4970	1.4975
74.032	0.585	25.384	1.4984	1.4978
75.766	2.837	21.397	1.4957	1.4963
75.614	0.584	23.802	1.4970	1.4964
76.335	1.802	21.863	1.4955	1.4957
75.838	0.638	23.524	1.4960	1.4962
76.140	1.197	22.664	1.4952	1.4959
77.299	3.101	19.600	1.4934	1.4949
77.479	2.916	19.605	1.4935	1.4947
77.017	1.711	21.272	1.4949	1.4951
78.521	0.554	20.925	1.4926	1.4938
<i>Panel B</i>				
59.719	14.943	25.338	1.5110	1.5105
62.859	6.464	30.676	1.5070	1.5077
70.898	6.190	22.912	1.4990	1.5006
60.710	5.438	33.852	1.5090	1.5096

Table 1 (continued)

SiO ₂ (mol%)	Al ₂ O ₃ (mol%)	Na ₂ O (mol%)	μ_D	μ_C
50.072	5.778	44.151	1.5190	1.5192
75.900	1.638	22.462	1.4973	1.4961
75.312	2.573	22.114	1.4967	1.4967
74.840	4.193	20.967	1.4952	1.4971
75.443	5.519	19.038	1.4944	1.4965

The compositions were originally given in wt%.

were collected from different sources [8–11]. The glasses have a wide variety of compositions. The concentrations range between about 0.6 and 15 mol% Al₂O₃, 19 and 47 mol% Na₂O and 50 and 79 mol% SiO₂. In spite of the great difference in the compositions, there is a unique dependence of μ_D on the SiO₂ content. It is concluded that the refractive index of such glasses is mainly determined by a sole factor, that is the SiO₂ concentration.

4. Discussion

Previous studies led to common remarks about silicate glasses. It was found that the volume V_u of any structural unit in alkali silicate [12,13] or alkaline earth silicate glasses [14,15] remains the same in mixed alkali silicate glasses [12]. The same was found in mixed alkali–alkaline earth silicate glasses [14,15] and in Al₂O₃-containing silicate glasses too [16]. It was also found that any of the structural units in alkali silicate or alkaline earth silicate glasses has a definite value for the differential refraction f_u [6,7]. The f_u values do not change in mixed alkali [6] and in mixed alkali–alkaline earth silicate glasses [7].

In the light of these general features it can be proposed that f_u factors in Na₂O–SiO₂ glasses would be the same in Na₂O–Al₂O₃–SiO₂ glasses. In Na₂O–SiO₂ glasses [6] it was found that $f_4 = 2.422 \times 10^{-24}$, $f_3 = 3.758 \times 10^{-24}$ and $f_2 = 5.035 \times 10^{-24}$. By using these values in Eqs. (4a) and (4b) a value of $(2.650 \pm 0.094) \times 10^{-24}$ could be obtained for f_{Al} . In turn these values of f_{Al} , f_4 , f_3 and f_2 can be used to calculate the refractive index for the glasses investigated. An acceptable agreement could be attained between the μ_D values and the calculated refractive index (μ_C). A better agreement

could be reached by making minor changes ($\leq 1\%$) in the values of f_{Al} , f_4 , f_3 and f_2 . The used parameters are $f_{\text{Al}} = 2.588 \times 10^{-24}$, $f_4 = 2.446 \times 10^{-24}$, $f_3 = 3.740 \times 10^{-24}$ and $f_2 = 5.040 \times 10^{-24}$.

There is another way to get the values of f_{Al} , f_4 , f_3 and f_2 . The fitting line in Fig. 1 has the equation

$$\mu_D = -8.854 \times 10^{-4}C + 1.5635, \quad (5)$$

where C is the concentration of SiO_2 in mol%. This equation can be rewritten in terms of the molar fraction of SiO_2 ($z = C/100$) as

$$\mu_D = -8.854 \times 10^{-2}z + 1.5635. \quad (5a)$$

On the other hand, Eq. (4a) can be reformed to

$$\begin{aligned} \mu = & z(2f_3 - f_4 - 2f_{\text{Al}})N_{\text{A}} \\ & + x(4f_3 - 4f_4 - 2f_{\text{Al}})N_{\text{A}} \\ & + (2f_4 - 2f_3 + 2f_{\text{Al}})N_{\text{A}}. \end{aligned} \quad (6)$$

By using the obtained differential refraction values ($f_{\text{Al}} = 2.588 \times 10^{-24}$, $f_4 = 2.446 \times 10^{-24}$ and $f_3 = 3.740 \times 10^{-24}$) Eq. (6) becomes

$$\mu = -8.732 \times 10^{-2}z - 7.226 \times 10^{-3}x + 1.5621. \quad (6a)$$

This relation reveals that the Na_2O content has a minor effect on μ . For the glass with the highest Na_2O content (about 47 mol%) the value for the second term is about 0.075 of that for the first one and it becomes lower with decreasing the Na_2O content. From Eq. (6a) it appears that the SiO_2 concentration is the variable that mainly determines the change in μ in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses. By comparing Eqs. (5a) and (6) it can be concluded that, for the first approximation, the first and the last terms are the same in both equations. It follows that the second term in Eq. (6) might equal to zero and therefore

$$(2f_3 - f_4 - 2f_{\text{Al}})N_{\text{A}} = -8.854 \times 10^{-2}, \quad (7)$$

$$(4f_3 - 4f_4 - 2f_{\text{Al}})N_{\text{A}} = 0 \quad (8)$$

and

$$(2f_4 - 2f_3 + 2f_{\text{Al}})N_{\text{A}} = 1.5635. \quad (9)$$

By solving Eqs. (7)–(9) one obtains $f_{\text{Al}} = 2.596 \times 10^{-24}$, $f_4 = 2.449 \times 10^{-24}$ and $f_3 = 3.747 \times 10^{-24}$. The values of f_{Al} and f_3 can be used for glasses having $0.5 < (\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 1$ (Eq. (4b)) to get $f_2 = 5.046 \times 10^{-24}$. The maximum difference between these values and the corresponding ones for $\text{Na}_2\text{O}-\text{SiO}_2$ glasses (Table 2) is about 1%. Fig. 1 and Table 1 show a quite good agreement between μ_D and the calculated refractive index (μ_C) for the glasses investigated. This is the best agreement that could be attained by using the parameters obtained from solving Eqs. (7)–(9). In a similar manner Eq. (4b) can be reformed to

$$\begin{aligned} \mu = & z(f_2 - 2f_{\text{Al}})N_{\text{A}} + x(4f_2 - 4f_3 - 2f_{\text{Al}})N_{\text{A}} \\ & + (2f_3 - 2f_2 + 2f_{\text{Al}})N_{\text{A}}. \end{aligned} \quad (10)$$

This relation can be obtained by considering that $y = (1 - x - z)$. Comparing Eqs. (5a) and (10) one obtains

$$(f_2 - 2f_{\text{Al}})N_{\text{A}} = -8.854 \times 10^{-2}, \quad (11)$$

$$(4f_2 - 4f_3 - 2f_{\text{Al}})N_{\text{A}} = 0 \quad (12)$$

and

$$(2f_3 - 2f_2 + 2f_{\text{Al}})N_{\text{A}} = 1.5635. \quad (13)$$

Eqs. (7)–(9) and (11)–(13) can be used to get useful relations between the f_u factors. From Eqs. (7) and (11) it can be indicated that

$$f_3 = \frac{f_4 + f_2}{2}. \quad (14)$$

This relation can also be obtained from Eqs. (8) and (12) or Eqs. (9) and (13). Eq. (14) is in agreement with previous conclusion [6,7] that each NBO, in the Q_n sites, has the same contribution to the refractive index. From Eqs. (8) and (12) it follows respectively that

Table 2

Differential refraction (f_u) for the structural units in $\text{Na}_2\text{O}-\text{SiO}_2$ [6] and $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses

Type of glass	f_4 (10^{-24})	f_3 (10^{-24})	f_2 (10^{-24})	f_{Al} (10^{-24})
$\text{Na}_2\text{O}-\text{SiO}_2$	2.422	3.758	5.031	—
$\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$	2.449	3.747	5.046	2.596

$$f_3 = f_4 + (f_{\text{AL}}/2) \quad (15)$$

and

$$f_2 = f_3 + (f_{\text{AL}}/2). \quad (16)$$

These relations correlate the differential refraction of the structural units in glass with each other. It can be proved that these relations would have a general form such as:

$$f_u = f_{u+1} + (f_{\text{AL}}/2), \quad (17)$$

where u ($0 \leq u \leq 3$) being the number of bridging oxygen ions per SiO_4 tetrahedron. Eq. (17) is in good agreement with the f_u values obtained for Na_2O – SiO_2 glasses [6].

The above conclusions seem to be puzzle. Eq. (17) reveals that f_u for a silicate unit is correlated with f_{Al} . In principle there is no expected dependence of f_u on f_{Al} because the former is not affected with the presence of different structural units in the matrix [6,7]. By considering the value of f_{Al} (2.596×10^{-24}) Eq. (17) can be rewritten as

$$f_u = f_4 + N (1.298 \times 10^{-24}). \quad (17a)$$

Here N represents the number of NBOs per SiO_4 tetrahedron. On the other hand there is a linear increase in f_u with increasing n in Na_2O – SiO_2 glasses [6]. The straight line has the equation

$$f_u = f_4 + N (1.3069 \times 10^{-24}). \quad (18)$$

The slope in Eqs. (17a) and (18) is the same. This is just a fortuitous agreement. It can be considered as the basic reason that Al_2O_3 content has no influence on μ for Na_2O – Al_2O_3 – SiO_2 glasses.

The above data for f_{Al} , f_4 , f_3 and f_2 can be used to calculate the refractive index for model Na_2O – Al_2O_3 – SiO_2 glasses. This would be useful to investigate the effect of each component on μ . Fig. 2 represents the dependence of μ_c on the Al_2O_3 content in glasses having constant concentrations of SiO_2 and $(\text{Al}_2\text{O}_3/\text{Na}_2\text{O}) \leq 1$. From the first sight it is noticed that Al_2O_3 has practically no effect on μ . Again it is shown that μ is greatly influenced by the SiO_2 content. Similar conclusions can be obtained from Fig. 3. It appears that also Na_2O has no effect on μ .

From a specific point of view the above conclusions seem strange. How would both Al_2O_3 and

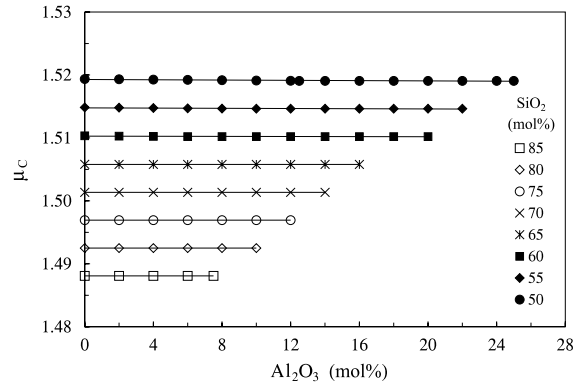


Fig. 2. Dependence on the Al_2O_3 content of the calculated refractive index (μ_c) for Na_2O – Al_2O_3 – SiO_2 glasses having constant SiO_2 concentration. The μ_c values were calculated by using Eqs. (4a) and (4b) and the f_u values given in Table 2.

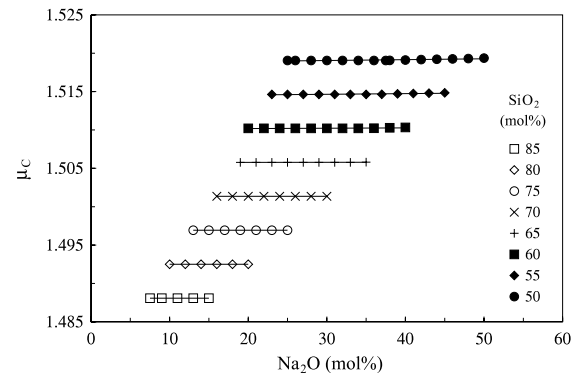


Fig. 3. Dependence on the Na_2O content of the calculated refractive index (μ_c) for Na_2O – Al_2O_3 – SiO_2 glasses having constant SiO_2 concentration. The μ_c values were calculated by using Eqs. (4a) and (4b) and the f_u values given in Table 2.

Na_2O have no effect on the refractive index of such glasses? The reason might certainly be correlated with the structural changes resulting from the change in the glass composition. Fig. 4 shows a linear dependence of $n_u f_u$ on the Al_2O_3 content in Na_2O – Al_2O_3 – SiO_2 glasses having 50 mol% SiO_2 . The contribution ($n_u f_u$) in the refractive index varies from a structural unit to another. As would be deduced from Eq. (2), for $0.5 < (\text{Na}_2\text{O} - \text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 1$, the increase in Al_2O_3 content is accompanied with a decrease in the number n_2 of Q_2 units. This results as an increase in n_3 . In Fig. 4 it is shown that the resultant effect of the units with

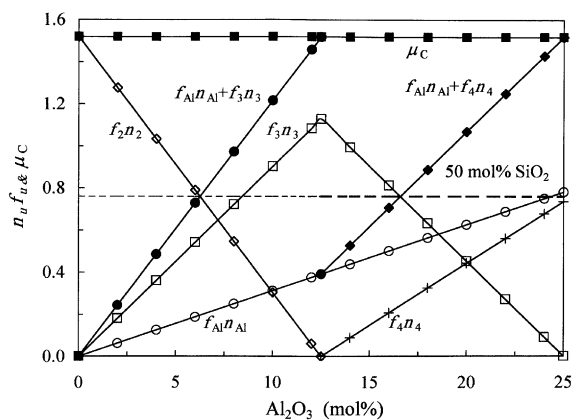


Fig. 4. Contribution in the refractive index ($n_u f_u$) of the structural units and the calculated refractive index (μ_c) as a function of the Al_2O_3 content in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses having 50 mol% SiO_2 . The $n_u f_u$ values can be calculated from Eqs. (4a) and (4b) and the f_u values given in Table 2.

increasing concentration ($f_{\text{Al}}n_{\text{Al}} + f_3n_3$) is compensated by the effect of the units with decreasing concentration (f_2n_2). For $(\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{SiO}_2 \leq 0.5$, i.e. for $\text{Al}_2\text{O}_3 > 12.5$ mol%, f_3n_3 decreases whereas $f_{\text{Al}}n_{\text{Al}}$ and f_4n_4 increase upon increasing the Al_2O_3 content. Also in this case the increase in ($f_{\text{Al}}n_{\text{Al}} + f_3n_3$) compensates the decrease in f_3n_3 . Therefore μ remains independent on the Al_2O_3 content. Such a behaviour is observed for any $\text{Al}_2\text{O}_3-\text{Na}_2\text{O}-\text{SiO}_2$ glass having $(\text{Al}_2\text{O}_3/\text{Na}_2\text{O}) \leq 1$, regardless of the Na_2O or Al_2O_3 content.

5. Conclusion

The refractive index of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses arises from contributions of the structural

units in glass. Each structural unit has a specific factor with which it contributes in the property. All the factors are constant and independent of the glass composition. The factors of the silicate units in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses are the same found for $\text{Na}_2\text{O}-\text{SiO}_2$ glasses. The refractive index depends only on the SiO_2 content. Each NBO makes the same contribution to the refractive index. The difference in the differential refraction of a Q_n and Q_{n-1} equals to $f_{\text{Al}}/2$. The results can be explained by considering the change in the concentration of the structural units and their contribution to the refractive index. The obtained factors are useful in calculating μ for glasses having $\text{Al}_2\text{O}_3/\text{Na}_2\text{O} \leq 1$.

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